

Program : Diploma in Chemical Engineering	
Course Code : 3073	Course Title: Inorganic Chemical Technology
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To interpret the industrial processes for the manufacturing of inorganic chemicals.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Basic Chemistry		Applied chemistry I	1
		Applied chemistry II	2
Unit operations and Unit process		Chemical Process Principles	2

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize chloralkali industries.	12	Understanding
CO2	Summarize the manufacturing of industrial acids.	16	Understanding
CO3	Summarise the manufacturing of fertilizers, cement and paint	20	Understanding
CO4	Summarize the manufacturing of glass, ceramics and refractories	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarize chloralkaly industries.		
M1.01	Illustrate the manufacture Sodium hydroxide	4	Understanding
M1.02	Illustrate the manufacture of sodium carbonate	4	Understanding
M1.03	Illustrate the manufacturing of lime	2	Understanding
M1.04	Illustrate the manufacturing of bleaching powder	2	Understanding
Contents: Outline Inorganic chemical industries – List the important inorganic chemical manufacturing industries in Kerala – Outline Chloralkali industries - Sodium hydroxide (caustic soda) - Raw materials - Membrane cell process - Requirements of membrane - Reactions – Flow diagram - Uses - Byproduct in caustic soda manufacture - Chlorine - Compression and liquefaction - Uses Sodium carbonate – Modified Solvay process - Raw materials - Source of raw materials – Reactions - Flow diagram - Preparation and purification of brine - Ammoniation of brine - Carbonation of ammoniated brine - Filtration – Calcination - Recovery of ammonia. Lime - Raw materials - Lime kiln – Flow diagram - Properties of lime – Uses Bleaching powder - Raw materials - Reaction - Industrial process - Flow diagram- Uses			
CO2	Summarize the manufacturing of industrial acids.		
M2.01	Illustrate the manufacturing of sulphuric acid	4	Understanding
M2.02	Illustrate the manufacturing of hydrochloric acid	4	Understanding
M2.03	Illustrate the manufacturing of nitric acid.	4	Understanding
M2.04	Illustrate the manufacturing of phosphoric acid.	3	Understanding
	Series Test – I	1	

Contents:

Sulphuric acid manufacturing process - DCDA process for sulfuric acid - Sources of raw material - Reactions - Melting and Burning of sulfur - Catalytic oxidation of SO_2 to SO_3 - Operation of multistage convertor with figure - absorption of SO_3 to form oleum - Dilution of oleum and cooling - Flow diagram - Uses.

Hydrochloric acid manufacturing - Synthesis process - Raw materials - Sources of raw material - Reaction - Operation of burner and absorber - Flow diagram - Uses

Nitric acid manufacturing - Ostwald's process or Ammonia oxidation process - Raw materials - Major reactions - Flow diagram - Catalyst poisoning - Concentration by H_2SO_4 - Uses

Phosphoric acid manufacturing - Wet process - Raw materials - Drying and grinding - Digestion - Filtration - Concentration - byproducts - Process flow diagram - Uses

CO3	Summarise the manufacturing of fertilizers, cement and paint		
M3.01	Illustrate the manufacturing of Nitrogenous fertilizers.	8	Understanding
M3.02	Illustrate the manufacturing of Phosphatic and Complex fertilizers.	3	Understanding
M3.03	Illustrate the manufacturing of Portland cement.	4	Understanding
M3.04	Illustrate the manufacturing of Paint and pigments	5	Understanding

Contents:

Nitrogenous fertilizers - List the commonly used nitrogenous fertilizers - Sources of hydrogen for ammonia synthesis - Production of hydrogen from naphtha - Steam reforming process - process flow diagram - Nitrogen from air (Linde- Frankl cycle) - Process flow diagram - Manufacturing of ammonia from synthesis gas - Illustrate ammonia reactor - CO_2 recovery - Applications - Urea manufacturing - Total recycle process - Reactions- Autoclave - Prilling tower - Byproducts - Uses - Ammonium chloride - Manufacturing process - Process flow diagram - Uses.

Phosphatic and Complex fertilizers - Manufacturing process - Superphosphate - Triple super phosphate - Di Ammonium Phosphate - NPK fertilizers - Uses

Cement - Raw materials of cement - Manufacturing of Portland cement - Methods for the manufacturing of Portland cement- Dry process - Process flow diagram - Crushing and grinding - Construction and working of rotary kiln - Grinding the clinker with gypsum - Storage and packing - Settling and hardening of cement.

Paint industries - Constituents of paints and their functions - Pigment - Thinners - Drying agents - Fillers - Plasticizers - Anti-skinning agents - Manufacturing of paint - flow diagram

Titanium dioxide - Raw materials - Manufacturing using chloride process - Flow diagram - Uses - Carbon black - Raw materials - Manufacturing using furnace process - Process flow diagram - Applications

CO4	Summarize the manufacturing of glass, ceramics and refractories		
M4.01	Illustrate the manufacturing of glass	3	Understanding
M4.02	Illustrate the manufacturing of ceramics	3	Understanding
M4.03	Illustrate the manufacture of refractory's	3	Understanding
M4.04	Comprehend the products of refractory's and their uses	2	Understanding
	Series Test – II	1	

Contents:

Glass - Raw materials - Chemical composition - Manufacturing Process – Batching – Melting –Forming - Pot furnace – Tank furnace – Diagram – Annealing - Types of glasses – Soda lime glass - Borosilicate glass - Special optical glass - Glass wool - Fused silica glass

Ceramics – Classification - Major components – Clays - Types of clay by their physical and refractory properties - Methods for fabrication of ceramic ware - Dry pressing - Hot pressing - Slip casting - Drying of ceramic ware - Firing of ceramic ware - List the types of ceramic products - Chemical stonewares

Refractory - Requisite of good refractory - Classification of refractory's - List the raw materials for the refractory's - Manufacture of refractory's - Preliminary treatment - Blending and mixing -Forming or moulding – Drying - Firing or burning

Types of refractory products - Fire clay refractory's – Uses - High-alumina refractory's – Uses -Magnesite refractory's – Uses - Carbon refractory's - Uses

Text / Reference

T/R	Book Title/Author
T1	Dryden's Outlines of Chemical Technology; Edited and revised by M. Gopala Rao and Marshall Sittig
R1	Shreve's Chemical Process Industries; G.T. Austin
R2	Chemical Process Technology; O.P. Gupta
R3	Encyclopedia of Chemical Technology ; Kirk-Othmer
R4	A Textbook of Engineering Chemistry; Dr. SS. Dhara and Dr. Umare S chand publishers.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/106/103106108/